

uses of derivative:

- control **credit risk**
- reduce **market risk**
- ↑ risk to ↑ **returns**
- **switch** asset allocations without disturbing underlying
[transition management]

Future's
options

Exchange

⊗ clearing house

- checks buy + sell orders match
- acts as party to every trade
- counterparty to both parties
- guarantees each side to the transaction

Advantages:

- removes counterparty risk
- indistinguishable contracts
∴ easier to close out

(C C FGH R)

Roles:

- counterparty to all trades
- Guarantor of all deals
- Registrar of deals
- Holder of deposited margin
- Facilitator of marking to market process

(Dot CCMMML)

Future's

standardised exchange
traded contract
between 2 parties to
exchange an A on a
set date at set price

price = closely related
to underlying asset

contract: (standard)

- units of trade
- how settlement price is to be determined
- exact details of underlying
- delivery date

Forward

non-standardised, privately
negotiated contract between
2 parties to trade on A
on a set date at a set price



OTC

< Forwards
Swaps

⊗ centrally cleared party (CCP)

Other derivative products:

⊗ Guaranteed Equity Products (G&EP)

- return linked to equity index
- min guaranteed return = 0

⊗ Structural notes

- non-standardised
- meet risk-return requirements
- embedded option + provides payment in pre-specified way

CHAPTER 2: DERIVATIVES

margin:

- collateral each party to
exchange traded derivative
deposits with CH

- cushion against potential
loss → defaulting agreement

⊗ Initial margin

⊗ Variation margin → controls
liquidity risk counterparty risk

→ excess of initial = withdrawn



← initial margin level
= maintenance margin

Future: both parties pay

Option: only writer pays

Types of margins:

- cash
- asset backed
- EFT / shares
- govern securities
- Gold

Definitions:

• Derivative	financial instrument whose value depends on value of another underlying asset
• Futures contract	standardised, exchange traded contract between 2 parties to exchange an asset on a set date in the future at set price
• Forward contract	non-standardised, ^{privately} negotiated contract between 2 parties to trade an asset on a set date at a set price
• Tick size	min price movements specified for a contract
• Margin	collateral each party to exchange traded derivative must deposit with CH
• Open Interest	# of contracts outstanding at one time
• Options	contracts agreed to between investors to trade an underlying security at a given date at a set price
• Bermudan Options	exercisable on specified days
• Asian Options	payoff based on average value of underlying asset over a period
• Swap	contract between 2 parties under which they agree to exchange a series of payments according to a pre-arranged formula